Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the battery cables. Refer to the original equipment manufacturer (OEM) service manual.
- Remove the cooling fan and spacers. Contact a Cummins® Authorized Repair Location.
- Remove the fan belt. Refer to Procedure 008-002 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-008-002-om.html)

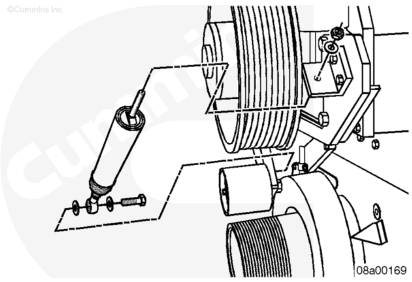
Remove

Linear Tensioner System

Remove the capscrews and washers from the lower clevis and idler arm.

Remove the nut and the capscrews from the threaded rod end of the tensioner at the upper anchor bracket.

Remove the fan belt tensioner.

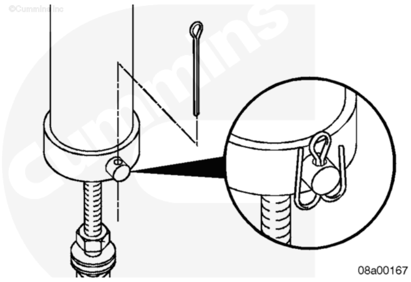


Disassemble

Linear Tensioner System

Remove the cotter pin from the clevis pin.

Remove the clevis pin from the control rod and belt tensioner housing and separate.



Remove the control rod (1) from the control rod retainer (2).

Remove and discard the o-ring (3) from the control rod retainer (2), if present.

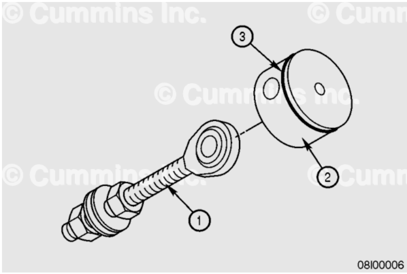
Remove the first nut from the control rod threaded end.

Remove the flat washers from the control rod threaded end.

Note : The number of washers may vary. Record the number of washers removed so the same number is reassembled, if required,

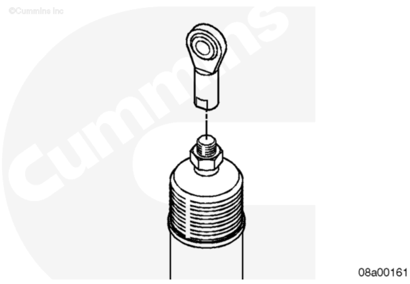
Remove the remaining nut from the control rod threaded end.

Note : Engines built on or after 01-Aug-2011 and ESN first 37250458 have an o-ring in a groove on the control rod retainer.



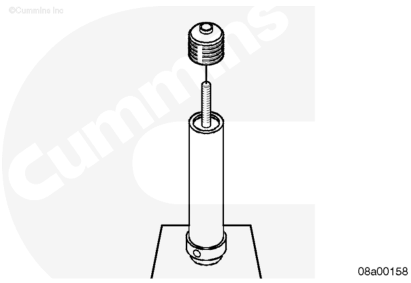
Remove the control rod from the capscrew threaded end.

Remove the nut holding the dust seal to the capscrew.



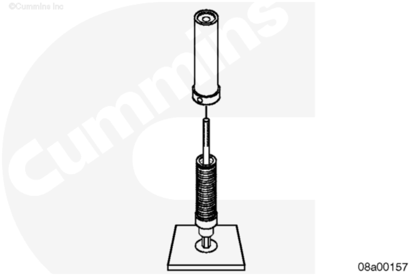
Cut the wire tie to remove the dust seal from the belt tensioner housing.

Remove the dust seal from the threaded end of the capscrew and belt tensioner housing.



Note : It is not necessary to use a build fixture to disassemble or assemble as shown in the illustration. This is an aid to clarify the steps in the procedure.

Remove the belt tensioner housing from the belt tensioner assembled parts.

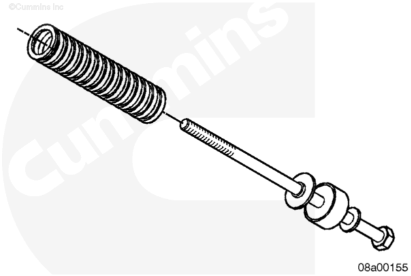


Remove the compression spring from the spring retainer.

Remove the spring retainer from the spring guide.

Remove the spring guide from the flat washer.

Remove the flat washer from the capscrew.

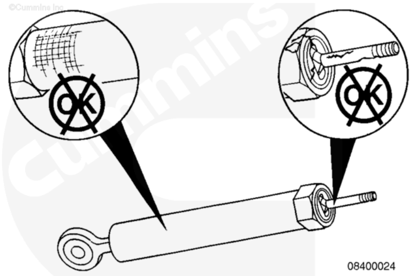


Inspect for Reuse

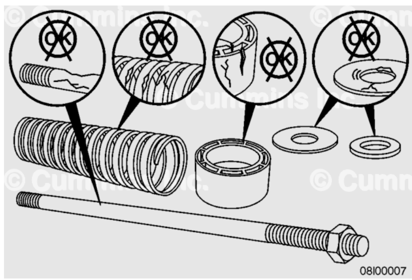
Linear Tensioner System

Inspect the tensioner assembly and ends for cracks or excessive wear.

If the tensioner assembly is cracked or worn, it **must** be replaced.



If the tensioner is disassembled, inspect the components for rust or corrosion. All rusted or corroded parts must be replaced.



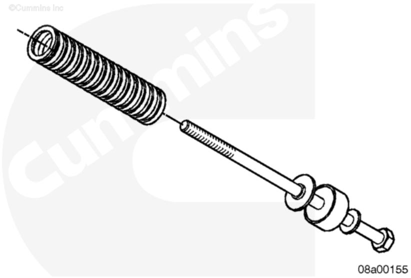
Assemble

Linear Tensioner System

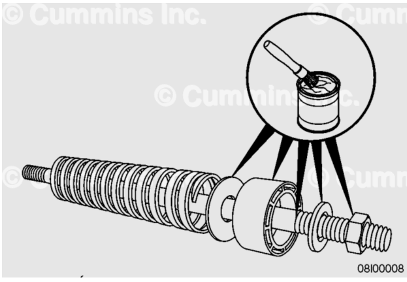
Install a washer on the capsrew so that the round side of the washer faces the capscrew head.

Install the spring guide with the recessed/indented side next to the flat side of the washer.

Install the spring retainer with the larger flat surface against the spring guide.

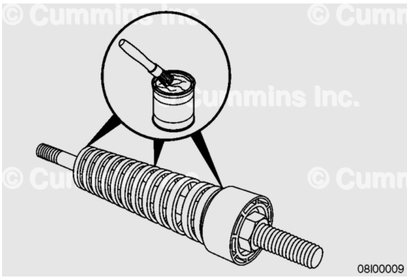


Use Lubriplate™ 105, Part Number 3163087, or equivalent, to grease the spring guide, washers, retainer, and top of the capscrew between the washers, retainer, and spring.

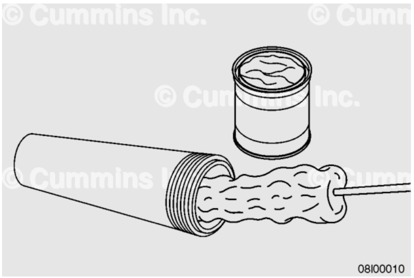


Install the compression spring against the spring retainer.

Use Lubriplate™ 105, Part Number 3163087, or equivalent, to grease the outside of the compression spring. Make sure to grease both contacting end surfaces.

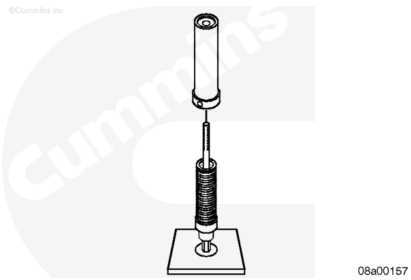


Use Lubriplate™ 105, Part Number 3163087, or equivalent, to grease the inside of the fan belt tensioner housing.



Note : It is not necessary to use a build fixture to diassemble or assemble as illustrated. This is an aid to clarify the steps in the procedure.

Install the belt tensioner housing over the assembled parts and allow the bottom of the housing to sit on the build fixture with the threads of the capscrews protruding out of the belt tensioner housing.

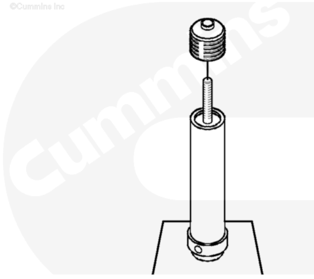


Slip the dust seal over the end of the capscrew and the edge of the belt tensioner housing.

Note : Use a ratchet to prevent the capscrew from moving while pushing the dust seal into place.

Use a wire tie and fasten around the dust seal where it overlaps the belt tensioner housing.

Pull the tie until tight and cut off the excess of the tie.



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Push the dust seal down over the threads of the capscrew and install the nut.

Back the nut against the control rod and tighten.

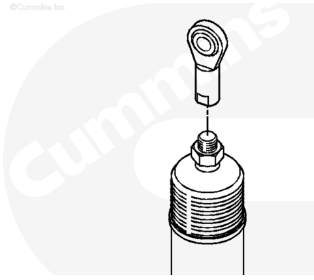
Use a 3/4-inch open-end wrench to tighten the nut.

Note : Make sure to tighten the nut just far enough to get clearance for the following part to be installed. Do not completely tighten at this time.

Install the control rod end on the threaded end of the capscrew.

Back the nut against the control rod and tighten.

Torque Value: 68 n•m [50 ft-lb]



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Apply Lubriplate™ 105, Part Number 3163087, or equivalent, to the o-ring groove (3) in the control rod retainer (2) and the spring guide.

Install the o-ring (3) on the control rod retainer (2) and install in the housing.

Install the nut on the control rod end (1).

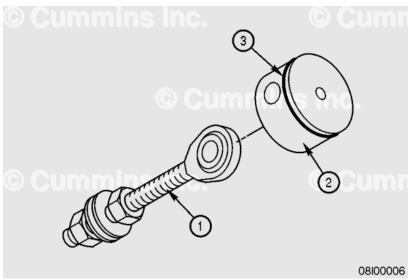
Install flat washers onto the control rod end.

Note : Make sure to install the same number of washers that were removed earlier.

Install a second nut onto the control rod end.

Note : Do not tighten completely. Leave the washers loose.

Install the assembled control rod end on the control rod retainer on the slotted end.

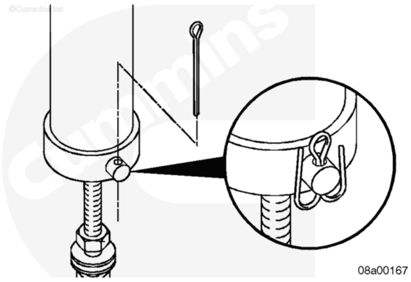


Use a non-metallic mallet, or equivalent, to install the retainer into the belt tensioner housing.

Align the holes and install the clevis pin to hold the parts together.

Use a cotter pin to secure the clevis pin to the housing.

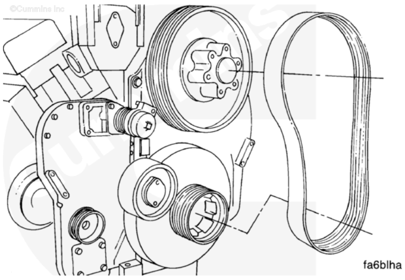
Bend the cotter pin around the clevis pin to keep it from falling out during operation.



Install

Note : The fan hub pulley and the fan belt are shown removed in the illustrations below for clarity.

Install the fan belt. Refer to Procedure 008-002 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-008-002-om.html)



Linear Tensioner System

Attach the lower clevis end of the tensioner to the idler arm with capscrews and washers.

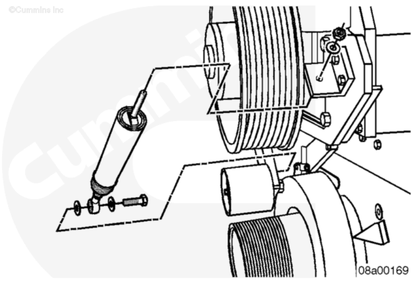
Tighten the capscrews.

Torque Value: 102 n•m [75 ft-lb]

Slide the threaded-rod end of the tensioner through the hole in the upper anchor bracket and secure it with a washer and nut.

Pull the belt tensioner until the idler pulley contacts the fan belt. Tighten the top nut hand-tight.

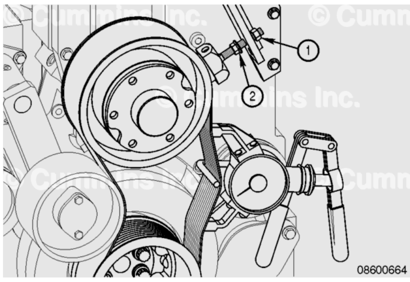
Note : Do not tighten the nut. It will be tightened later to adjust the belt tension.



Adjust

Linear Tensioner System

Adjust the fan belt and check belt tension. Refer to Procedure 008-002 in Section 4. (/qs3/pubsys2/xml/en/procedures/20/20-008-002-om.html)



Finishing Steps

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the cooling fan and spacers. Contact a Cummins Authorized Repair Location.
- Connect the battery cables. Refer to the OEM service manual.

Last Modified: 05-Nov-2013